CS & IT ENGINEERING

Operating System

CPU Scheduling

Lecture -3

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Recap of Previous Lecture

Topic SJF Scheduling

Topics to be Covered

Topic SRTF Scheduling

Topic LJF Scheduling

Topic LRTF Scheduling

Topic : SJF (Shortest Job First)

Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time
P1	0	5			
P2	1	3			
P3	2	4			
P4	4	2			
P5	5	1			
P6	6	5			

P1	P5	P4	P2	P3	P6	
0	5	6	8	11	15	20

Topic : SJF (Shortest Job First)

Advantages:

1. Minimum average waiting time among non-preemptive scheduling
2. Better throughput in continuous execution
   → no. of processes executed in one unit of time

Disadvantages:

1. No practical implementation because Burst time is not known in advance
2. No option of Preemption
3. Longer Processes may suffer from starvation
   → indefinite wait

Topic : SRTF (Shortest Remaining Time First)

Scheduling Criteria: smallest BT first | Tie breaker ⇒ FCFS

Type of Algorithm: Preemptive

Preemption ⇒ only when a new arriving process has small BT than remaining time of current running.

Topic : SRTF (Shortest ~~Job~~ First)

Remaining time

Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time
P1	0	4	7	7	3
P2	1	2	3	2	0
P3	2	1	4	2	1

P1	P2	P3	P1

0 1 ~~2~~ 3 4 7

no. of context switch = 3

Topic : SRTF (Shortest Job First)

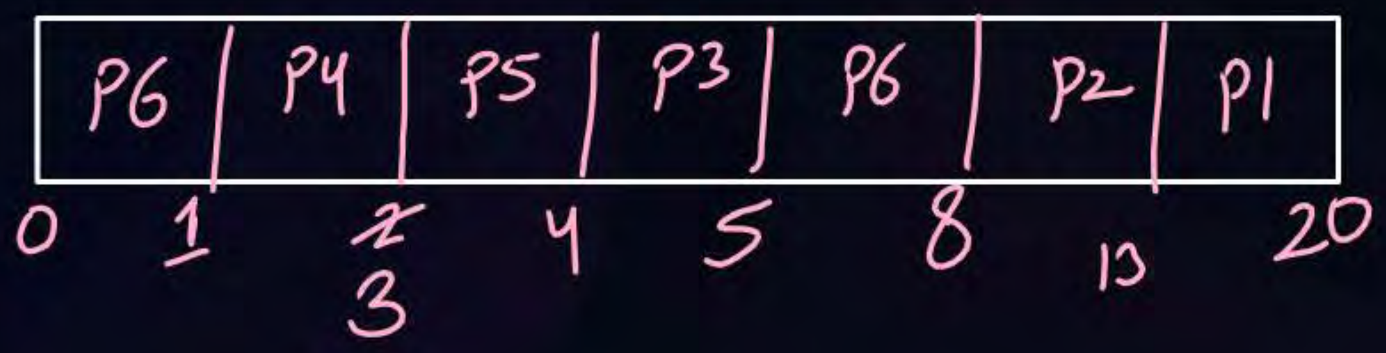
Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time
P1	0	6	10	10	4
P2	1	5	19	18	13
P3	2	1	3	1	0
P4	3	2	5	2	0
P5	4	1	6	2	1
P6	5	4	14	9	5

P1	P3	P4	P5	P1	P6	P2

0 ~~1~~ 2 3 ~~4~~ 5 6 10 14 19

Topic : SRTF (Shortest Remaining Time First)

Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time
P1	4	7			
P2	5	5			
P3	3	1			
P4	1	2			
P5	2	1			
P6	0	4			



Topic : SRTF (Shortest Remaining Time First)

Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time
P1	0	9	22		
P2	1	6	14		
P3	2	4	9		
P4	3	2	5		
P5	6	1	7		

P1	P2	P3	P4	P3	P5	P3	P2	P1	
0	1	2	3	5	6	7	9	14	22

Topic : SRTF (Shortest Remaining Time First)

Advantages:

and min avg TAT

1. Minimum average waiting time among all scheduling algorithm
2. Better throughput in continue run

Disadvantages:

1. No practical implementation because Burst time is not known in advance
2. Longer Processes may suffer from starvation

[NAT]

#Q. Response time of processes in non-preemptive scheduling algorithms are equal to waiting time of processes?

True or False

Justify your answer with appropriate explanation.

Topic : LJF (Longest Job First)

Scheduling Criteria: Largest BT first | Tie breaker = FCFS

Type of Algorithm: Non-preemptive

⇒ suffers from convoy effect

⇒ small processes may suffer from starvation

Topic : LJF (Longest Job First)

Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time
P1	0	6			
P2	0	3			
P3	0	4			
P4	0	2			

P1	P3	P2	P4

0	6	10	13	15

Topic : LJF (Longest Job First)

Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time
P1	0	2			
P2	1	3			
P3	2	4			
P4	3	2			

P1	P3	P2	P4

0 2 6 9 11

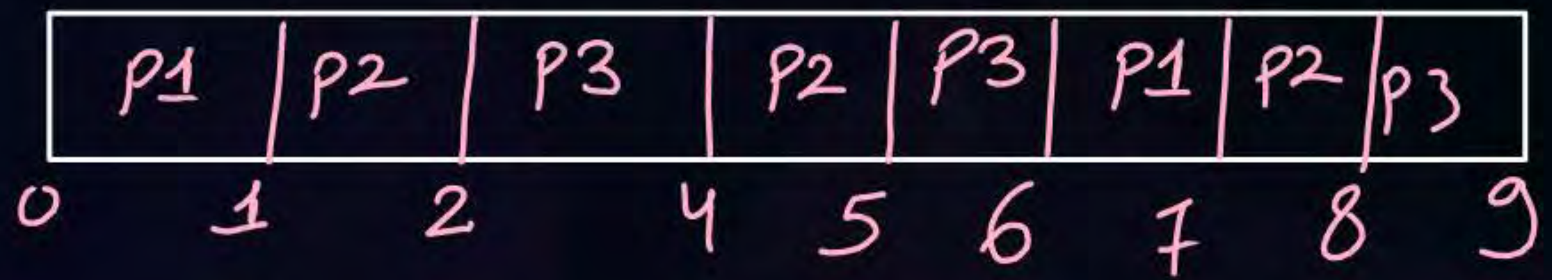
Topic : LRTF (Longest Remaining Time First)

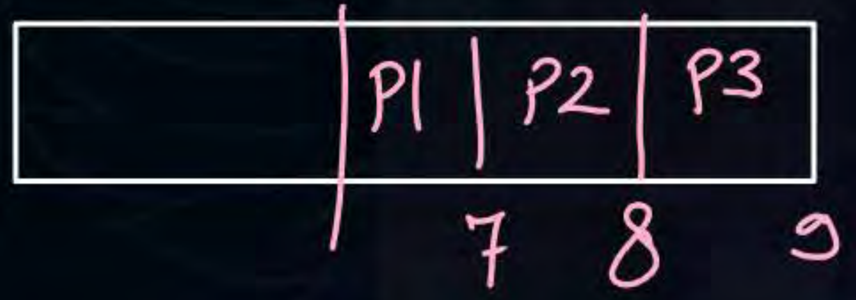
Scheduling Criteria: Largest BT process first | Tie breaker = FCFS

Type of Algorithm: Preemptive

Topic : LRTF (Longest Remaining Time First)

Process	Arrival Time	Burst Time	Completion Time	Turnaround Time	Waiting Time
P1	0	2	7		
P2	1	3	8		
P3	2	4	9		





[MCQ]

GATE – PYQ

#Q. Consider three processes (process id 0, 1, 2 respectively) with compute time bursts 2, 4 and 8 time units. All processes arrive at time zero. Consider the longest remaining time first (LRTF) scheduling algorithm. In LRTF ties are broken by giving priority to the process with the lowest process id. The average turn around time is:

A ✓ 13 units

B 14 units

C 15 units

D 16 units

	P0	P1	P2
	12	13	14

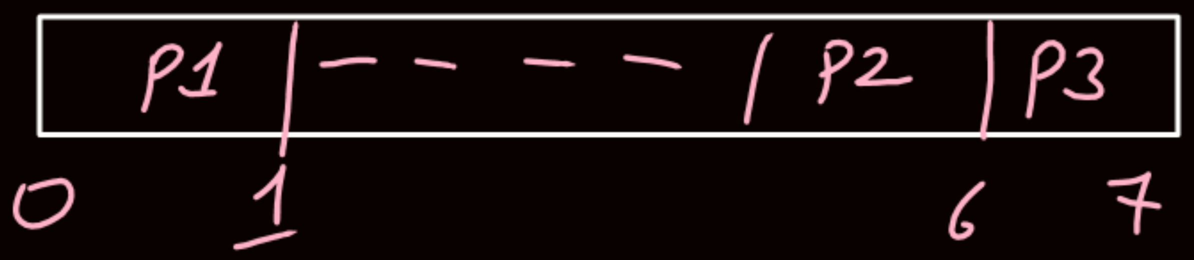
		TAT
P0	12	12
P1	13	13
P2	14	14

avg = 13

Ques)

	AT	BT	
P1	0	1	
P2	1	2	1
P3	2	4	

before P2 and P3 arrive,
P1 completed.



Topic : HRRN (Highest Response Ratio Next)

Objective: Not only favors short jobs but decreases the WT of longer jobs.

no starvation

Topic : HRRN (Highest Response Ratio Next)

Scheduling Criteria: Highest Response ratio | Tie breaker ⇒ SJF

Type of Algorithm: Non-preemptive

$$\text{Resonse Ratio} = \frac{W + BT}{BT}$$

W = Wait Time

BT = Service/Burst Time

Topic : HRRN (Highest Response Ratio Next)

Process	Arrival Time	Burst Time
P1	0	3
P2	2	6
P3	4	4
P4	6	5
P5	8	2

P1	P2	P3	P5	P4

0 3 9 13 15 20

At time 9:-

$RR(P3) = \frac{5+4}{4} = 2.25$ (Highest)

$RR(P4) = \frac{3+5}{5} = 1.6$

$RR(P5) = \frac{1+2}{2} = 1.5$

At time 13:-

$RR(P4) = \frac{7+5}{5} = 2.4$

$RR(P5) = \frac{5+2}{2} = 3.5$ (Highest)

2 mins Summary

Topic SRTF Scheduling

Topic LJF Scheduling

Topic LRTF Scheduling

Happy Learning

THANK - YOU